

AI-Powered Trek Risk & Route Intelligence System

A Geospatial Data Science Project Brief

1. Ideation — What the project actually is

You're building a decision-support system for trekkers in hill terrain that answers four questions before and during a trek:

1. **Where is the terrain dangerous right now?** (landslide/rockfall susceptibility + real-time trigger conditions)
2. **How dangerous, exactly?** (a graded risk score, not just yes/no)
3. **If the planned path becomes unsafe, what's the next-best route?** (alternative path routing)
4. **If someone is stuck in a bad zone, what should they do?** (survival guidance, tied to the specific hazard type detected)

The core insight that makes this a *geospatial* problem and not just a weather app: landslide/trail danger is a function of **static terrain properties** (slope, aspect, curvature, soil, geology, land cover) combined with **dynamic triggers** (rainfall intensity, antecedent soil moisture, temperature/freezing-thaw, seismicity). Static factors tell you *where* danger concentrates; dynamic factors tell you *when* a static hotspot activates. Your model needs both layers, and the fusion of the two is the actual novel contribution — not any single technique in isolation.

This kind of system sits precisely at the intersection of two well-established but rarely-combined research areas: **landslide susceptibility mapping (LSM)** (a mature GIS/ML field) and **least-cost-path / network routing** (a mature GIScience field). Combining them into a live "safe trek routing" tool is a legitimate, publishable-scope idea — recent literature does exactly the LSM half at scale but almost nothing pairs it with live routing for recreational trekkers specifically. That's your niche.

2. Research Areas to Learn (and why each matters here)

2.1 Hazard Analysis (Landslide Susceptibility Mapping — LSM)

This is the academic name for "where will a landslide happen." The standard methodology, repeated across dozens of papers:

- **Conditioning factors** are extracted as raster layers: slope, aspect, curvature (plan/profile), elevation, TWI (topographic wetness index), distance to drainage/roads/faults, lithology, land use/NDVI, rainfall, and soil type.

- **Landslide inventory** (historical landslide point/polygon locations) is the *labelled ground truth* — without this, you have no way to train or validate a model. Points are typically split ~70/30 train/test.
- **Models used in current literature:** classic statistical approaches (Frequency Ratio, Analytical Hierarchy Process/AHP, Weight-of-Evidence) versus ML approaches (Random Forest, XGBoost, SVM, CatBoost, Neural Networks) versus modern ensemble/stacking approaches (CNN+RF, CNN+CatBoost, stacking ensembles with a meta-learner). Recent papers (2025) increasingly report ensembles/stacking beating any single model, evaluated with ROC-AUC.
- **A recent development worth knowing:** a 2026 paper explored replacing hand-picked conditioning factors with Google's AlphaEarth satellite embeddings feeding a deep learning model — a sign the field is moving toward learned features rather than manually engineered ones. You don't need this for a first version, but it's a legitimate "future scope" item.
- India-specific: the **Geological Survey of India (GSI)** runs the **National Landslide Susceptibility Mapping (NLSM)** programme — 1:50,000 scale maps now covering ~4.3 lakh km² across the Himalayan belt, NE India, and Western Ghats, with ~91,000 historical landslides catalogued (33,904 field-validated). This is a goldmine because it's exactly the labelled inventory + susceptibility ground truth that most papers have to build by hand.

Key papers to read (search these titles):

- *Integrated Machine Learning Approaches for Landslide Susceptibility Mapping Along the Pakistan–China Karakoram Highway* (2025) — good example of CNN/RF/CatBoost stacking methodology, directly transferable to a mountain highway/trail context.
- *Assessment of Landslide Susceptibility using Geospatial Techniques* — Kalimpong district, West Bengal (Raut et al., 2025, *Geological Journal*) — **this is in your own state**, so its factor list and GIS workflow is the most directly reusable reference for you.
- *National-scale landslide susceptibility and risk mapping of India using a hybrid data-driven approach* (*Scientific Reports*, 2025) — gives you the national baseline (AHP, Frequency Ratio, Yule's Coefficient) and shows what "high susceptibility" looks like at 90m resolution across India — useful for sanity-checking your own model's output.
- *Comparative assessment of machine learning models for landslide susceptibility mapping: a focus on validation and accuracy* (*Natural Hazards*, 2025) — focused specifically on how validation is done properly, which you'll need for your own methodology section.
- *Frictional timescales and the impact of climate change-driven extreme weather on rainfall-triggered landslides in Mizoram, NE India* (2025, arXiv) — a smaller-scale, rainfall-trigger-focused study close to your rainfall-trigger use case.

2.2 Terrain Analysis (Geomorphometry)

This is the toolkit you'll actually use inside QGIS to turn a raw elevation raster into the conditioning-factor layers above:

- **Slope, Aspect, Curvature** — the three basic morphometric parameters; SAGA's "Slope, Aspect, Curvature" tool computes all of them (plus profile/plan curvature) from a DEM in one batch operation.
- **TWI (Topographic Wetness Index)** and **flow accumulation** — needed to model where water concentrates, a major landslide trigger; requires "filling sinks" in the DEM first (SAGA's "Fill Sinks (Wang & Liu)" or GRASS `r.fill.dir`) so flow paths are continuous.
- **TPI (Topographic Position Index)** and **terrain ruggedness** — help distinguish ridge/valley/slope positions and rate how rugged a path segment is (relevant for "difficulty level").
- **Hillshade** — for visualization, not analysis, but useful in your final maps.
- **Least-cost path (LCP) analysis** — this is the actual routing algorithm behind "alternative path": you build a *cost surface* raster (a grid where each cell's value = difficulty/danger of crossing it), and then compute the cheapest path across it between two points. This is standard in QGIS + GRASS (`r.cost` + `r.drain`) and is exactly the tool used for trail/access-route planning in the wild (see the **TopoTrail** QGIS plugin below).

2.3 Route Optimization / Network Analysis

Once you have a cost surface, two complementary approaches exist:

- **Raster-based least-cost path** (continuous terrain, good for off-trail or when no existing path network exists) — GRASS `r.cost/r.walk`.
- **Graph-based routing** (when you already have a trail network as a graph of nodes/edges) — this is where **NetworkX** comes in: you weight each edge by a combined cost (distance + slope penalty + live danger score), and re-run Dijkstra/A* whenever conditions change, to produce the "alternative safest path" in real time. This is the natural fit for your "extreme condition → reroute" feature, since trekking routes are mostly known trail networks, not open terrain.

2.4 Weather Integration

You need both **forecast** (what's coming) and **historical/reanalysis** (what triggered past landslides, for training) data. **Open-Meteo** is the standout free option: no API key required, global coverage, and — critically for you — provides **soil moisture and soil temperature at multiple depths** alongside standard rainfall/temperature/wind, both as live forecasts and as an 80+ year ERA5/ERA5-Land historical reanalysis archive. That reanalysis archive is exactly what you'd use to reconstruct rainfall conditions on days when historical landslides occurred in your inventory — a standard part of the LSM "trigger factor" methodology.

2.5 Soil Data

For static soil properties (texture, depth, permeability — relevant to slope stability, separate from moisture): **ISRIC SoilGrids** is the standard free global soil property dataset. For India specifically, cross-check against **Bhukosh** (GSI's portal), which also carries lithology/geology layers alongside its landslide inventory.

3. Data Collection Sources (all free/open)

Data type	Source	Notes
DEM (elevation)	Bhuvan/NRSC CartoDEM (30m, India-specific, ISRO) or SRTM (30m/90m, global, via USGS EarthExplorer/OpenTopography)	CartoDEM is India-specific and slightly better accuracy for Indian terrain; SRTM is easier to script/download programmatically
Landslide inventory (ground truth)	GSI Bhukosh (bhukosh.gsi.gov.in) — part of the National Landslide Susceptibility Mapping (NLSM) programme	Free download, point/polygon shapefiles, huge historical database (~91,000 events, ~34,000 field-validated)
Weather (forecast + historical)	Open-Meteo API (open-meteo.com) — no key needed, JSON, includes soil moisture/temperature	Use the Forecast API for live conditions, Historical Weather API (ERA5/ERA5-Land) for training-time reconstruction of past rainfall
Soil properties	ISRIC SoilGrids (soilgrids.org)	Global 250m resolution, REST API + WCS available
Land cover / NDVI	Sentinel-2 via Copernicus Open Access Hub, or Bhuvan's LULC layers	Needed as a conditioning factor (vegetation stabilizes slopes)
Geology/lithology	GSI Bhukosh geology layers	Rock type strongly affects slope failure mechanics
Trail/path network	OpenStreetMap (via Overpass API or <code>osmnx</code> in Python)	Gives you the existing graph of trekking paths/roads to route over
Seismicity (secondary trigger)	National Center for Seismology (India) or USGS earthquake catalog	Optional — relevant since some Himalayan landslides are seismically triggered

4. Tools You'll Learn Alongside the Project

QGIS (desktop GIS — your main workspace)

- **Core raster tools:** Raster Calculator, Slope/Aspect (native + SAGA/GRASS backends), Fill Sinks, Flow Accumulation.
- **SAGA GIS** (bundled with QGIS via Processing Toolbox): best for detailed morphometry (slope/aspect/curvature/TWI/TPI in one batch run) and hydrology (watershed delineation, flow accumulation).
- **GRASS GIS** (also bundled): best for `r.cost` (cost-surface generation) and `r.walk/r.drain` (least-cost path extraction) — this is your route-alternative engine.

- **Useful plugin: TopoTrail** — a QGIS plugin purpose-built for trail-suitability analysis using DEM, slope, curvature, and least-cost routing. Worth installing early since it's essentially a reference implementation of half your pipeline.
- **MAS Spatial Analysis Tool** (plugin) — a consolidated 45+ tool DEM/hydrology/geomorphometry toolkit if you want faster batch processing than clicking through SAGA/GRASS individually.

GeoPandas / Python geospatial stack

- `geopandas` — vector data (shapefiles, GeoJSON) as a Pandas-like dataframe; your landslide inventory and trail network will live here.
 - `rasterio` — reading/writing/sampling raster layers (DEM, slope, weather grids) in Python.
 - `rasterstats` — zonal statistics: e.g. "what's the mean slope within 100m of this trail segment."
 - `osmnx` — pulls trail/road networks from OpenStreetMap directly as a `networkx` graph — this is likely your fastest path to a working route graph.
 - `networkx` — graph algorithms (Dijkstra/A*) for the actual route optimization once edges are weighted by danger score.
 - `scikit-learn` / `xgboost` + `shap` — the ML model for susceptibility prediction and its explainability layer (SHAP tells you *why* a location is rated high-risk — useful both for validation and for generating human-readable "danger reason" text for the survival-guide feature).
-

5. Step-by-Step Implementation Plan

Phase 0 — Scoping (1–2 weeks) Pick ONE real trek region to prototype on (don't try to build this generically for "all of India" — pick e.g. a Darjeeling/Sikkim/Kalimpong trail cluster near you, since GSI/Bhukosh data density is good there and it lets you validate against real, recent landslide events). Define the trail network boundary and download DEM + inventory for just that region first.

Phase 1 — Data Acquisition (1–2 weeks)

- Download DEM (CartoDEM or SRTM) for your study area.
- Download landslide inventory points from Bhukosh for the same area.
- Pull the trail/road network via `osmnx`.
- Set up Open-Meteo API calls (historical archive for past dates matching your landslide inventory; forecast endpoint for live use later).
- Pull SoilGrids data for the area (texture, depth).

Phase 2 — Terrain Feature Engineering in QGIS (2–3 weeks)

- Fill sinks on DEM (SAGA "Fill Sinks (Wang & Liu)").
- Derive slope, aspect, plan/profile curvature (SAGA "Slope, Aspect, Curvature" batch tool).

- Derive flow accumulation and TWI (SAGA hydrology tools or GRASS `r.watershed`).
- Derive TPI/ruggedness.
- Rasterize/vectorize distance-to-drainage, distance-to-road, land cover, lithology into aligned raster layers (same resolution/extent/CRS — this alignment step is where most beginners lose time, so budget for it).

Phase 3 — Susceptibility Model (2–3 weeks)

- Sample all conditioning-factor rasters at landslide points (positive class) and an equal number of random non-landslide points (negative class) — this is the standard LSM sampling strategy.
- Train XGBoost classifier on the resulting tabular dataset; hold out 30% for testing.
- Evaluate with ROC-AUC, precision/recall (imbalanced classes matter here).
- Use SHAP to identify which factors drive predictions — cross-check against literature (slope and TWI/rainfall usually dominate) as a sanity check on your own data pipeline.
- Apply the trained model back over the full raster grid to produce a continuous susceptibility map for the region.

Phase 4 — Dynamic Risk Layer (1–2 weeks)

- Pull live/forecast rainfall + soil moisture from Open-Meteo for the region.
- Combine static susceptibility score with a rainfall-trigger multiplier (e.g., $\text{susceptibility} \times \text{normalized recent rainfall intensity}$) to get a *time-varying* danger score per grid cell — this is what turns a static hazard map into something usable "right now."

Phase 5 — Route Optimization (2–3 weeks)

- Build a cost surface from the dynamic danger raster (danger + slope penalty + distance).
- Option A (raster): GRASS `r.cost/r.walk` for off-trail least-cost path.
- Option B (graph, recommended as primary): weight each edge of the `osmnx/networkx` trail graph using average danger-raster value along that edge (via `rasterstats zonal stats`), then run Dijkstra/A* for shortest-safe-path and a re-route function that recomputes if a segment's danger crosses a threshold.

Phase 6 — Difficulty Scoring

- Combine slope, elevation gain, ruggedness, and trail length into a difficulty index (this is a simpler weighted-sum model, not ML — keep it interpretable since "difficulty" is subjective and users will want to understand it).

Phase 7 — Survival Guide Layer

- This is a rules/lookup layer, not ML: map hazard type (rainfall-triggered landslide zone, steep rockfall zone, river-crossing zone, extreme cold zone) to a corresponding pre-written guidance snippet. Trigger the relevant snippet based on which factor dominates the SHAP explanation for that grid cell.

Phase 8 — Validation

- **Spatial:** hold-out test set ROC-AUC/AUC-PR against known landslide points (standard LSM validation).
- **Temporal:** for a handful of historical landslide dates in your inventory, reconstruct the weather conditions via Open-Meteo's historical archive and check whether your dynamic risk layer would have flagged that location as high-risk on that date — this is your strongest, most defensible validation story since it ties both the static and dynamic halves of the model to real events.
- **Field-level sanity check** (optional but valuable): compare your model's high-susceptibility zones against GSI's own published NLSM susceptibility map for the same region — if they broadly agree, that's strong external validation.

Phase 9 — Interface

- A simple web map (Leaflet/Folium, or a Streamlit app) showing: base trail network, susceptibility heatmap, live danger overlay, computed alternative route, and a difficulty/survival-guide panel per selected trail.

6. Validation Strategy (summary)

1. Train/test split (70/30) on landslide inventory — standard ML validation.
2. ROC-AUC / precision-recall on held-out points.
3. SHAP-based factor importance cross-checked against published literature for similar Himalayan terrain (slope, TWI, rainfall should dominate — if your model says otherwise, that's a data-quality red flag, not a discovery).
4. Historical event backtesting using Open-Meteo's reanalysis archive against known landslide dates.
5. Cross-check against GSI's own NLSM susceptibility map for your chosen region, where available.

7. Forward Scopes (what to propose as "future work")

- Swap hand-engineered conditioning factors for learned satellite embeddings (e.g., Google AlphaEarth) — an active 2026 research direction.
- Add a **seismic trigger layer** (many Himalayan landslides are earthquake-aggravated, not just rainfall-triggered).
- Move from a static/rule-based difficulty score to a **crowdsourced trekker feedback loop** (actual completion times, reported conditions) to continuously recalibrate difficulty and danger scores.
- **Early-warning push notifications:** tie your dynamic risk layer to a threshold-crossing alert system (SMS/app push) for trekkers already registered on a route — this mirrors what GSI itself is piloting at a national scale.

- Extend from a single-region prototype to a multi-region system using GSI's now-available national NLSM coverage as a base layer, with your dynamic/routing layer added on top region-by-region.
- Mobile offline mode: cache the susceptibility map + trail graph for offline use in low-connectivity mountain areas (a genuine practical gap in existing trekking apps).

8. Key References (for your literature review section)

- Nohani, E. et al. — *Integrated Machine Learning Approaches for Landslide Susceptibility Mapping Along the Karakoram Highway*, 2025 (MDPI Land).
 - Raut, S., Dutta, D., Bera, D., Samanta, R. — *Assessment of Landslide Susceptibility using Geospatial Techniques: A Comparative Evaluation of Machine Learning and Statistical Models*, Kalimpong district, West Bengal — *Geological Journal*, 2025.
 - *National-scale landslide susceptibility and risk mapping of India using a hybrid data-driven approach* — *Scientific Reports*, 2025.
 - Abdelkader, M.M., Csámer, Á. — *Comparative assessment of machine learning models for landslide susceptibility mapping: a focus on validation and accuracy* — *Natural Hazards*, 2025.
 - *Frictional timescales and the impact of climate change-driven extreme weather on rainfall-triggered landslides in Mizoram, NE India* — arXiv, 2025.
 - *GIS-Based Landslide Susceptibility Mapping with a Blended Ensemble Model* — Sentani, Papua, Indonesia — *MDPI Geosciences*, 2025.
 - *Testing least cost path (LCP) models for travel time and kilocalorie expenditure: Implications for landscape genomics* — useful methodological reference specifically for trekker-route cost modelling (travel time/energy expenditure as path cost, rather than just distance).
 - GSI National Landslide Susceptibility Mapping (NLSM) programme — official government reference for India-wide susceptibility baseline and inventory (bhukosh.gsi.gov.in).
-

This brief is meant as your working roadmap — expect Phases 1–3 to take the bulk of your learning time since terrain feature engineering in QGIS is genuinely the steepest part of the curve. Everything after Phase 3 reuses skills you'll already have